# 8.1.1. MULTIPLICATION OF 2 SQ. MATRICES

**ALGORITHM**

**Step 1:** Start

**Step 2:** Input Dimension (n)

**Step 3:** A=[]

Create an empty list A

Set loop a in range 0 to n:

Input a row of integers and append to A

**Step 4:** B=[]

Create an empty list B

Set loop b in range 0 to n:

Input a row of integers and append to B

**Step 5:** Result =[]

Create an empty list result

Set loop r in range 0 to n:

Append a list of n zeros to result

**Step 6:** Multiplication of Matrix

Set loop i in range 0 to n:

Set loop j in range 0 to n :

Set loop k in range 0 to n :

result[i][j] += (A[i][k] * B[k][j])

Step 7: Print Resultant Matrix

Set loop i in range $0$ to $n$:

Set loop j in range $0$ to $n$:

If j is the last index :

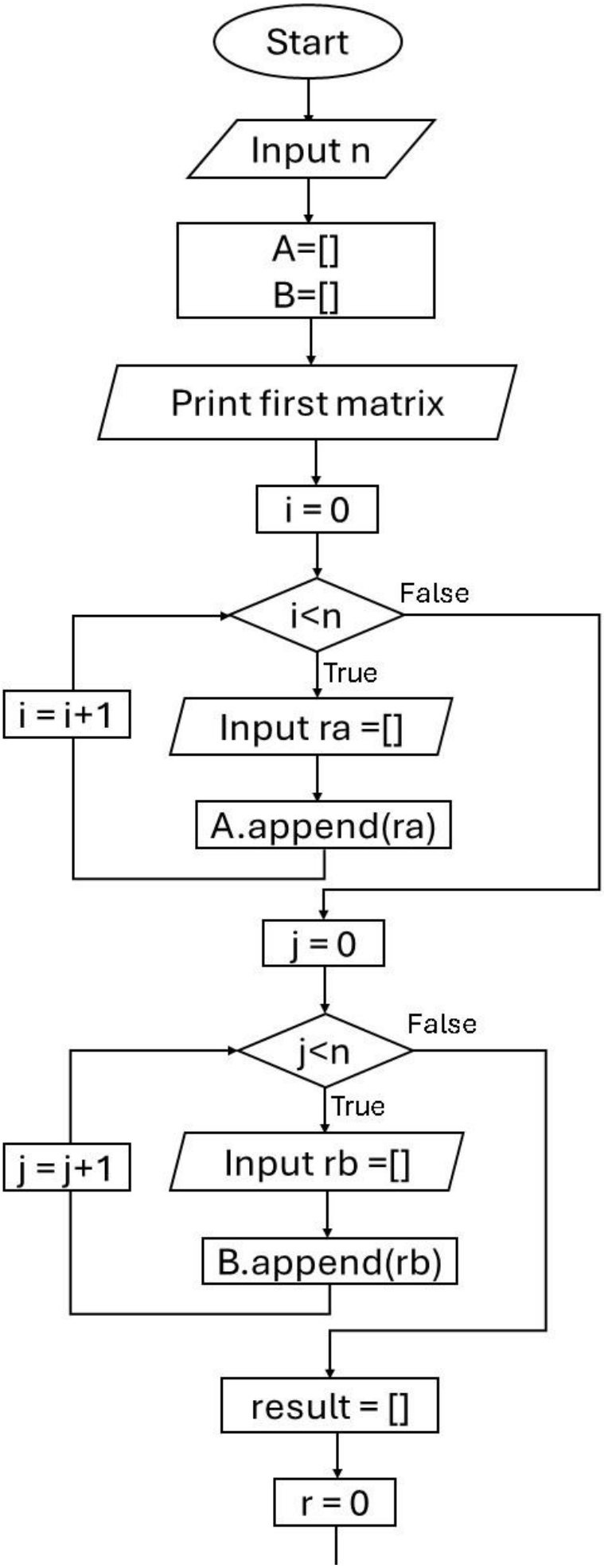
Print result[i][j] (end="")

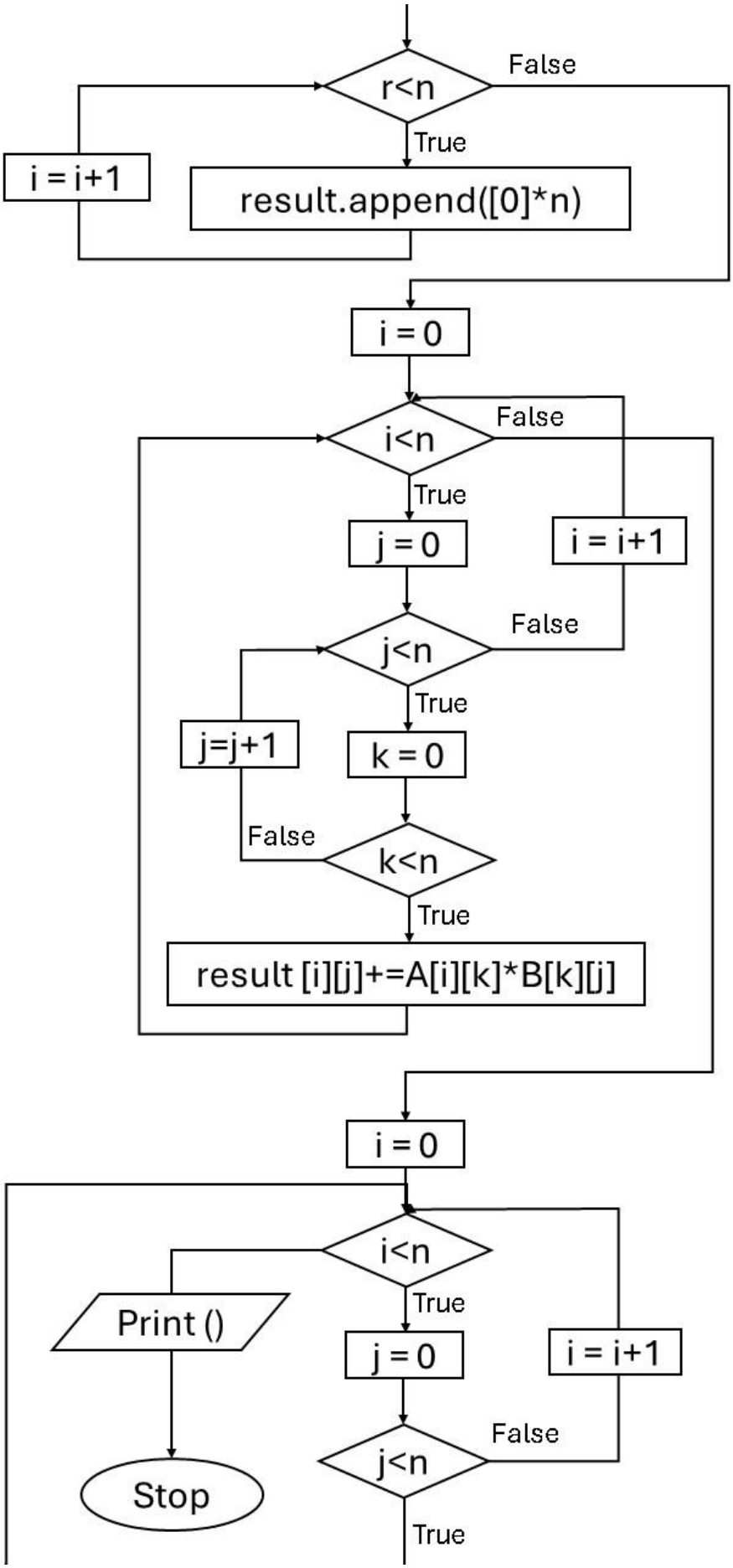
Else:

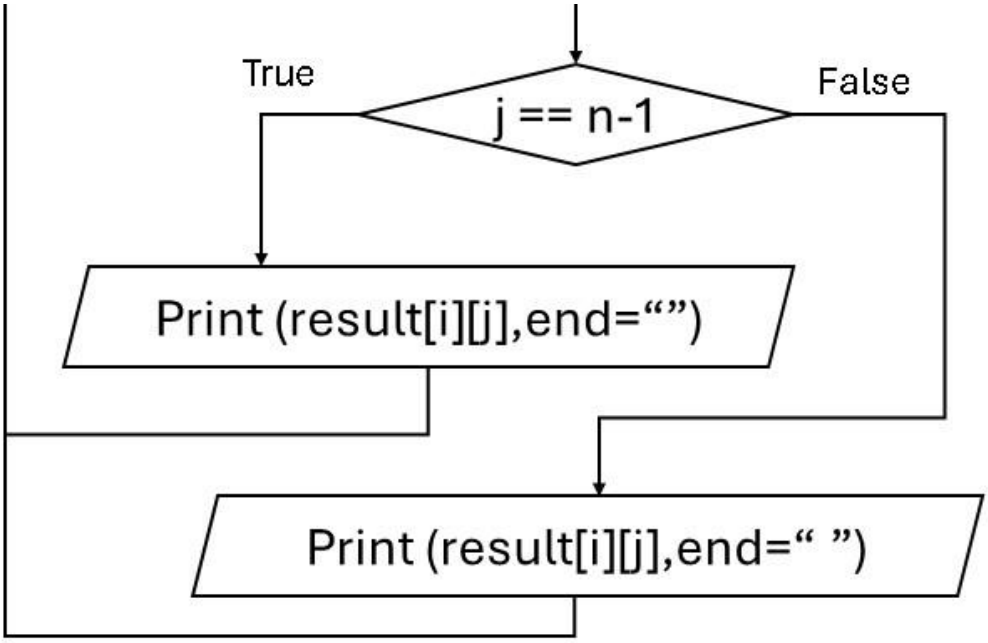
Print result[i][j](end=" ")

Print a new line.

**Step 8:** Stop


Start
Input n
A=[]
B=[]
Print first matrix
i = 0
i<n
False
True
i = i+1
Input ra =[]
A.append(ra)
j = 0
j<n
False
True
j = j+1
Input rb =[]
B.append(rb)
result = []
r = 0


r<n
False
True
i = i+1
result.append([0]*n)
i = 0
i<n
False
True
j = 0
i = i+1
j<n
False
True
j=j+1
k = 0
False
k<n
True
result [i][j]+=A[i][k]*B[k][j]
i = 0
i<n
True
Print ()
j = 0
i = i+1
j<n
False
Stop
True

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**8.1.1. Multiplication of Two Square Matrices** 07:58

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

**Input Format:**

- The first line prints the "dimension: " and reads an integer $n$, representing the dimensions of the square matrices ($n \times n$).
- The second line prints: "first matrix:" and then reads $n$ lines, each containing $n$ integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads $n$ lines, each containing $n$ integers separated by spaces, representing the rows of the second matrix.

**Output Format:**

- Print "Resultant Matrix:" followed by $n$ lines, each containing $n$ integers separated by spaces, representing the product of the two matrices.

Sample Test Cases +

Explorer

Multiplica... Submit

```
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    row = list(map(int, input().split()))
    A.append(row)

print("second matrix:")
B = []
for i in range(n):
```

Average time 0.016 s 16.00 ms

Maximum time 0.020 s 20.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 19 ms Debug

| Expected output | Actual output |
|---|---|
| dimension: 2 | dimension: 2 |
| first matrix: | first matrix: |
| 1 2 | 1 2 |
| 3 4 | 3 4 |
| second matrix: | second matrix: |
| 5 6 | 5 6 |

Terminal Test cases

Debugger

Prev Reset Submit Next